

Nick Cardamone

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EDUCATION

- **Masters of Applied Science, Electrical and Computer Engineering** Ottawa Canada
University of Ottawa Sept 2023 - Present
Thesis: Transmission Line Autorouter using Shape Synthesis with Dr. McNamara and Dr. Acimovic
Specialization: Electromagnetics, CEM, Microwave and RF Design
Courses: Phase Locked Loops and RX Synchronizers, Nonlinear Microwave Devices and Modeling, Signal and Power Integrity in High-Speed Designs, Antenna Engineering, Method of Moments
- **Bachelor of Applied Science, Electrical Engineering** Ottawa, Canada
University of Ottawa; Magna Cum Laude Sept 2018 - April 2023
Courses: Microwave Circuits, Antennas, Electronics Design, Modern Controls and Signal Processing, Wireless and Optical Communication Systems, Embedded Systems and Scripting

SKILLS SUMMARY

- **Electronics:** PCB and Circuit Design, Automated Testing, Test Equipment Expertise
- **Embedded Systems:** ARM, STM32, FPGA (Lattice, Xilinx, Altera), Embedded Linux (PetaLinux), RTOS
- **Programming:** C/C++, Python, MATLAB, VHDL, Rust, LaTeX, Git, Bash
- **Software:** Altium Designer, APD, AWR Microwave Office, HFSS, Fusion 360, SPICE

EXPERIENCE

- **Skyworks** Ottawa, ON
RFIC Design COOP Sept 2024 - Present
 - **RF Testing:** Validated power amplifier IC prototypes and managed automated test benches.
 - **RF Simulation:** Used HFSS and ADS to analyze amplifier and package performance.
 - **Design:** Designed harmonic filters for MCM substrates in APD and HFSS using a PyAEDT scripting approach.
 - **Troubleshooting:** Debugged automated test software to support tapeout deadlines.
 - **IC Rework:** Performed soldering and rework on packages using GaAs, SOI dies and 01005 components.
- **University of Ottawa** Ottawa, ON
Teaching Assistant Sept 2023 - April 2024
 - **Lab Instruction:** Taught Circuit Theory 1 labs covering soldering, AC/DC circuits, and lab instrumentation.
 - **Tutorials:** Delivered weekly sessions on circuit analysis techniques.
 - **Project Management:** Guided student projects, ensuring structured development and progress.
- **Arkalumen** Ottawa, ON
Sr Hardware Design COOP Jan 2021 - April 2023
 - **PCB Design:** Developed and tested IoT lighting control systems, including DC-DC converters.
 - **Firmware:** Implemented embedded software on nRF52, MSP430, and PIC microcontrollers.
 - **Fabrication:** Produced PCB layouts, BOMs, and prototypes for manufacturing.
- **Sino-can Energy** Tweed, ON
COOP Student May 2020 - Aug 2020
 - **Design:** Developed electronics hardware for an autonomous greenhouse.
 - **Fabrication:** Constructed greenhouse structures using 6061 aluminum.
 - **Installation:** Installed PV solar systems at multiple locations in Canada.

PROJECTS

- **USB-C Galvanic Isolator:** Designed a USB 3.2 isolator with 100W power isolation and 20 Gb/s data throughput.
- **Microwave Power Amplifier:** Designed a 3 GHz Class A RF amplifier for S-parameter optimization.
- **Patch Antenna Array:** Designed a 5 GHz microstrip antenna array and characterized it in an anechoic chamber.
- **Neural Network Buck Converter:** Developed an NN-based controller in Simulink to regulate DC-DC power conversion.
- **Reconfigurable FPGA-Based Filter:** Designed an FPGA-based programmable filter for use in a PLL.
- **Python Instrument Drivers:** Developing automated test instrument control drivers using SCPI, PyVISA and ChatGPT.

HOBBIES AND INTERESTS

- **Electronics:** Prototyping circuits, RF systems, and power electronics.
- **3D Printing:** Building enclosures and mechanical parts for electronics projects.
- **Music:** 10+ years playing drums and piano in jazz and concert bands.